

Brian W. Smith

PRINCIPAL SITE RELIABILITY ENGINEER · PLATFORM ENGINEERING · DEVELOPER TOOLING · OBSERVABILITY AT SCALE

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25+	100+	SLO	7,000+
YEARS IN PRODUCTION	KUBERNETES CLUSTERS OPERATED AT SCALE	SLI / ERROR BUDGET FRAMEWORKS	MICROSERVICES OPERATED AT SCALE

SUMMARY

Principal SRE at Red Hat with 25+ years of production systems experience, specializing in developer tooling, internal platform automation, and observability infrastructure that eliminates engineering toil. Deep hands-on expertise in the Grafana observability stack (Thanos, Prometheus, Alertmanager, Grafana). Deep expertise in operating environments with 100+ clusters and 7,000+ microservices, CI/CD pipeline engineering, Terraform IaC, and Grafana/Prometheus/Thanos observability across 100+ customer Kubernetes clusters. Security-first mindset built on zero trust principles, identity-based access controls, and FIPS-validated cryptography across AWS GovCloud environments. Currently driving platform reliability and AI-powered operational automation across the ROSA (Red Hat OpenShift Service on AWS) Regional Platform.

TECHNICAL SKILLS

- | | | |
|--|---|--------------------------------------|
| ▸ HashiCorp Vault / Secrets Management | ▸ CI/CD / Developer Tooling (Tekton, Prow, Jenkins) | ▸ Terraform / Infrastructure as Code |
| ▸ Prometheus / Grafana / Thanos / Alertmanager | ▸ Python (familiar) / Go (proficient) / Bash | ▸ OpenShift / Kubernetes (CKA) |
| ▸ AWS / Cloud-Native Infrastructure | ▸ Zero Trust / Identity-Based Networking | ▸ Keycloak / IAM / SSO |
| ▸ GitOps / ArgoCD / Helm / Kustomize | ▸ Incident Response / SLI/SLO / Error Budgets | ▸ Claude AI / MCP / Agent SDK |
| ▸ Distributed Systems / Large-Scale Operations | ▸ FedRAMP High ATO | |

Red Hat

November 2021 – Present

PRINCIPAL SRE · PRINCIPAL SRE TEAM LEAD · SENIOR SRE – ROSA REGIONAL PLATFORM

Principal SRE (January 2026 – Present)

- Automated FedRAMP compliance evaluation with a Claude Sonnet sub-agent pipeline — 6-step workflow (clone → classify → analyze → report → Jira → PR) covering ~800 NIST 800-53 controls, with Trivy CVE integration auto-downgrading verified controls on critical findings. github.com/theautoroboto/compliance-agent
- Validated Red Hat's FIPS-enabled Go toolchain (BoringCrypto) against AWS EKS nodes running Bottlerocket OS in FIPS kernel mode — using negative-control testing to confirm binary-level crypto enforcement aligned with kernel-level FIPS state via `/proc/sys/crypto/fips_enabled`; collaborated with Red Hat cryptographic maintainers to resolve compliance gaps. github.com/theautoroboto/fips-validation
- Implemented 13+ NIST 800-53 Rev 5 control families for FedRAMP Moderate compliance across the ROSA Regional Platform, mapping each against HCP Terraform modules to produce a gap-closed control inventory.
- Deployed a Thanos/Prometheus/Alertmanager/Grafana observability suite via parameterized IaC, enabling single-variable regional rollouts across all ROSA environments.
- Designed and deployed a Secret Management framework via AWS Secrets Manager and Vault sidecar injection, enforcing least-privilege IAM policies per service and eliminating plaintext secrets from environment variables across all services.
- Built a Python/DynamoDB Sandbox account leasing system provisioning scoped IAM roles and SCPs with auto-reclaim, cutting environment setup from hours to minutes.
- Architected a 3-tier AWS CodePipeline GitOps hierarchy with a meta-pipeline provisioner dynamically managing per-cluster pipelines from Git commits, paired with ArgoCD ApplicationSet Matrix Generators for zero-config multi-cluster delivery.
- Eliminated VPC peering and Transit Gateway coupling by engineering zero-trust cross-cluster messaging over AWS IoT Core MQTT; deployed ECS Fargate bootstrap-then-handoff for fully private EKS initialization; authored a 17-module reusable Terraform library.

Principal SRE - Team Lead (January 2025 – January 2026)

- Led FedRAMP CONMON remediation across a fleet of clusters and resolved SRE emergency access blockers by integrating backplane-cli with FedRAMP-compliant auth protocols.
- Architected AWS PrivateLink for ROSA HCP across all GovCloud regions with cross-account VPC Peering.
- Led the migration of 8 ROSA clusters across 2 GovCloud regions from classic to Hosted Control Plane (HCP) architecture, driving cross-functional coordination across multiple Red Hat engineering and product teams to deliver a successful FedRAMP compliance assessment.
- Developed and delivered 10+ technical training sessions on ROSA HCP to SRE teams across all GovCloud regions, driving large-scale adoption of a newly launched managed control-plane architecture and reducing time-to-productivity for engineers onboarding to the platform.
- Deployed Dynatrace Managed Service across FedRAMP environments with OneAgent log forwarding; engineered a Keycloak → Splunk log pipeline for centralized audit trails and real-time authentication security monitoring; authored SCR architecture diagrams and ADRs for External DNS Operator and HCP authorization.
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Senior SRE Functional Lead (December 2023 – January 2025) · Senior SRE (November 2021 – December 2023)

- Led a team of 9 SREs; defined and drove agile epics aligned with business priorities — CVE mitigation, annual FedRAMP assessments, Significant Change Requests, and operational toil reduction.
- Architected FedRAMP High GovCloud network fabric: Transit Gateway hub-and-spoke, per-tier VPC segmentation, PrivateLink, cross-account peering, VPC Flow Logs, and WAF — securing the ROSA platform foundation.
- Deployed Red Hat build of Keycloak (RHBK) to production, authored FedRAMP Boundary Architecture Diagrams, and engineered real-time synthetic availability monitoring for the identity platform.

- Automated AWS Config, GuardDuty, and SecurityHub via Terraform to establish the FedRAMP security baseline; authored the SSP Access Control validation epic for the ROSA Regional Platform.
- Led GoAlert deployment as the FedRAMP on-call platform from PoC through VPCE network architecture to production, replacing PagerDuty.
- Built the greenfield FedRAMP High GovCloud platform on AWS; established Terraform as the IaC provisioning foundation; led ATO workstreams including annual FedRAMP assessment support.

OneStream Software

February 2021 – November 2021

FIRST DEVOPS ENGINEER

- As the organization's first DevOps engineer, migrated 70+ repositories from TFS to Azure DevOps and established CI/CD foundations including SonarQube SAST, GitFlow branching, and Azure Pipelines.

Immuta

August 2020 – February 2021

CLOUD ENGINEER

- Deployed and configured Immuta's data security platform for enterprise customers across Databricks, Athena, Snowflake, and Redshift, applying Kubernetes expertise to accelerate onboarding across data warehouse environments.

Fiserv

March 2017 – August 2020

SENIOR SYSTEM ENGINEER LEAD – CLOUD PLATFORM

- Designed and built 16 Pivotal Cloud Foundry foundations supporting 7,000+ microservices; deployed Kubernetes into Azure using Pivotal Kubernetes Service (PKS/TKGI).
- Led Fiserv's PaaS deployment to Azure and authored the enterprise Terraform infrastructure; presented the hybrid-cloud Terraform architecture at SpringOne 2019.
- Built a self-service Kubernetes cluster provisioning platform (web form → Git → pipeline → Helm-deployed Prometheus/Grafana stack + expiration cleanup); designed centralized SLI dashboards across 14 PAS foundations using Grafana, AlertManager, and Prometheus.
- Formed and trained the Cloud Platform team, establishing agile delivery practices across the enterprise Kubernetes and PaaS program.

Earlier Roles

1998 – 2008

GRANGE INSURANCE (SENIOR SYSTEM ENGINEER) • NETJETS (SR. NETWORK ADMIN) • BMW FINANCIAL (CRM ADMIN) • CARDINAL HEALTH (SR. MESSAGING ENGINEER) • WD PARTNERS (IT ADMIN)

- 25+ years of progressive infrastructure experience spanning messaging, networking, virtualisation, enterprise AD/Exchange, and automation — from first ISP support role (BellSouth.net) through senior engineering leadership.

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)	CKA-2000-008308-0100
Azure DevOps Engineer Expert (AZ-400)	December 2020
Azure Solutions Architect Expert (AZ-303/304)	October 2020
Azure Administrator Associate (AZ-104)	October 2020

EDUCATION & NOTABLE

Franklin University — Computer Science	Columbus, OH
Eagle Scout — Order of the Arrow	Boy Scouts of America
SpringOne 2019 Speaker	Pivotal Platform / Azure
Open Source Contributor	github.com/theautoroboto